

WIRESPACES / BOARD 1 REV A

Four MCU bench prototype: two shared CAN–FD buses, multidrop UART and SAM ring.
Active PC USB power, 500 mA configured budget. Independent FTDI controls the main 3.3 V rail.

USB input and power

02 USB input and power

File: power.kicad_sch

USB bridge and VCP

03 USB bridge and VCP

File: usb_bridge.kicad_sch

Gateway STM32G474

04 Gateway STM32G474

File: gateway.kicad_sch

SAM0

05 SAM0

File: sam0.kicad_sch

SAM1

06 SAM1

File: sam1.kicad_sch

SAM2

07 SAM2

File: sam2.kicad_sch

CAN A

08 CAN A

File: can_a.kicad_sch

CAN B

09 CAN B

File: can_b.kicad_sch

UART and external terminals

10 UART and external terminals

File: interfaces.kicad_sch

Test access

11 Test access

File: access.kicad_sch

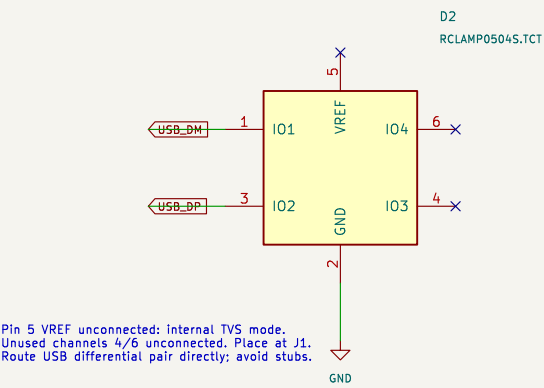
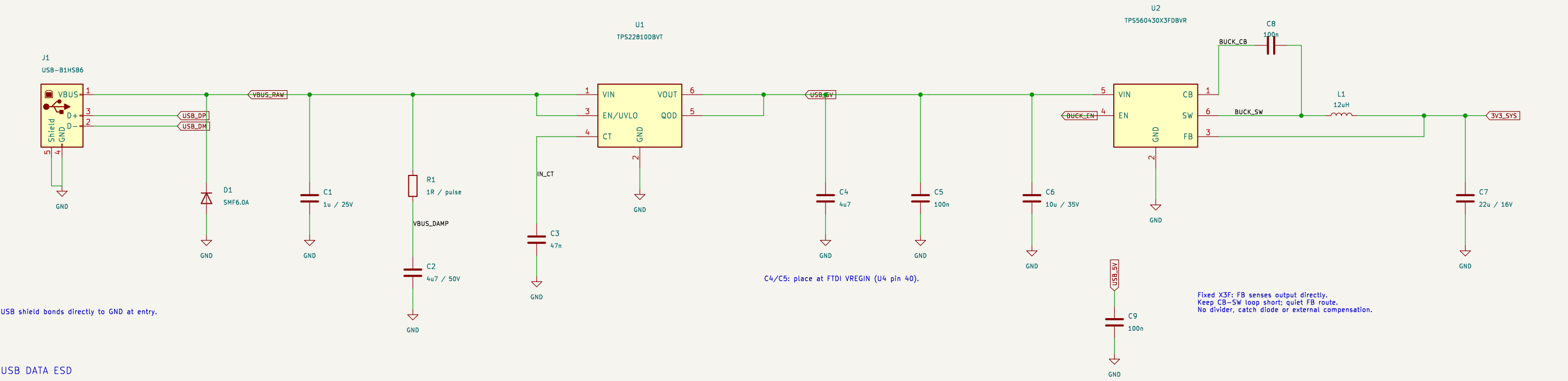
CAN A/B: GW + SAM0 + SAM1 + SAM2, eight PHYs.
UART_MD: four receivers; one LV125 channel per transmitter.
SAM ring: SAM0 -> SAM1 -> SAM2 -> SAM0.
Each MCU: SWD, private debug UART, SYNC/TRIG and EVENT0/1.

Named global nets join sheets; local analog nets stay within their sheet.
Review PDF and checks describe draft–capture evidence, ERC, layout/DRC
and hardware bring–up are separate stages. No PCB layout yet.

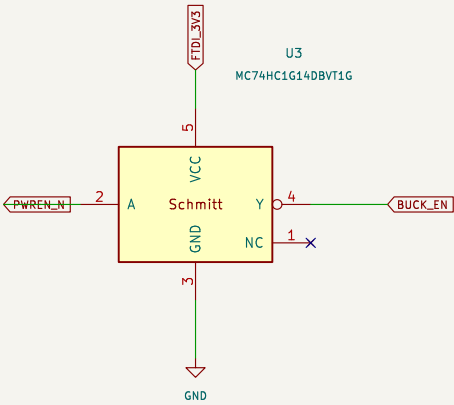
Rev A complete schematic draft – no PCB layout		
WireSpaces		
Sheet: /		
File: board1.kicad_sch		
Title: Board 1 – Rev A system overview		
Size: A3	Date: 2026–09–07	Rev: A–draft
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4	Id: 1/11	

USB-B INPUT AND ATTACHMENT RAMP

3.3 V MAIN BUCK



USB CONFIGURATION / SUSPEND CONTROL



PWREN_N low: main rail ON, VCP buffers enabled. High / floating: main rail OFF (ADR-040).



Rev A complete schematic draft – no PCB layout

WireSpaces

Sheet: /USB input and power/
File: power.kicad_sch

Title: Board 1 – USB input and main power

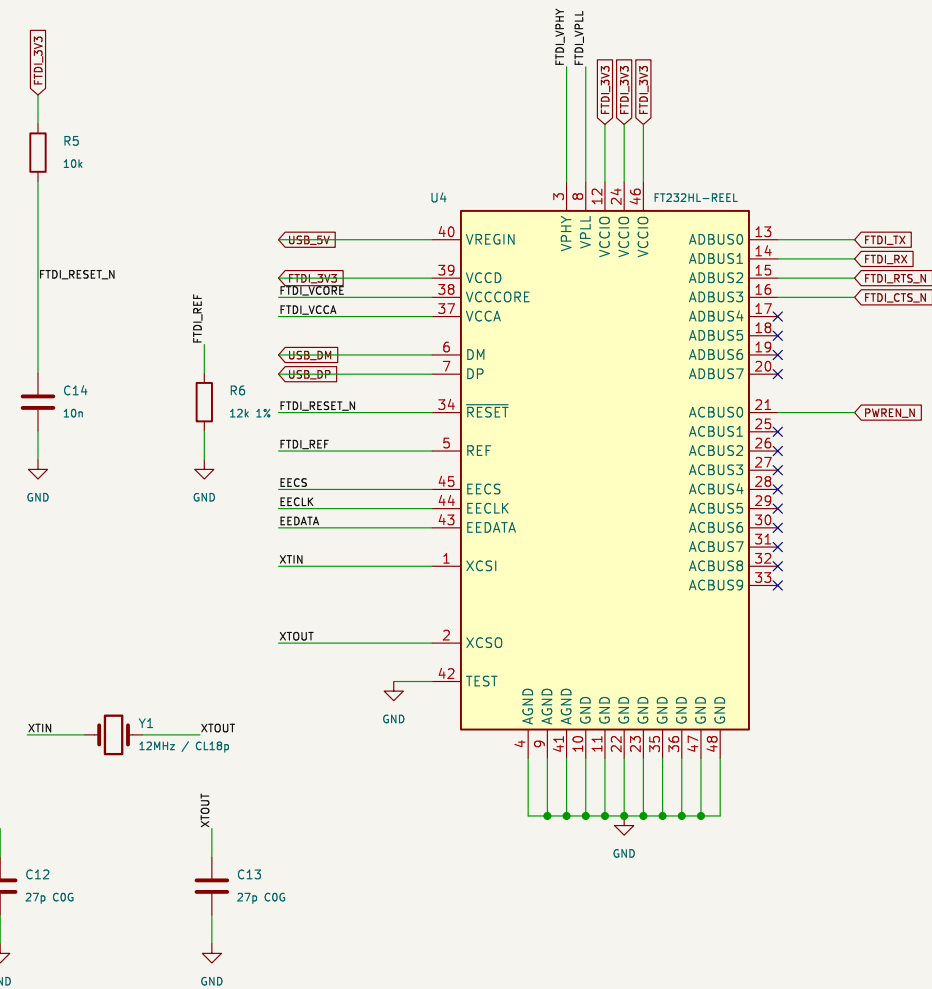
Size: A3 Date: 2026-09-07
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: A-draft
Id: 2/11

INDEPENDENT USB BRIDGE

CONFIGURATION EEPROM

VCP POWER-OFF ISOLATION

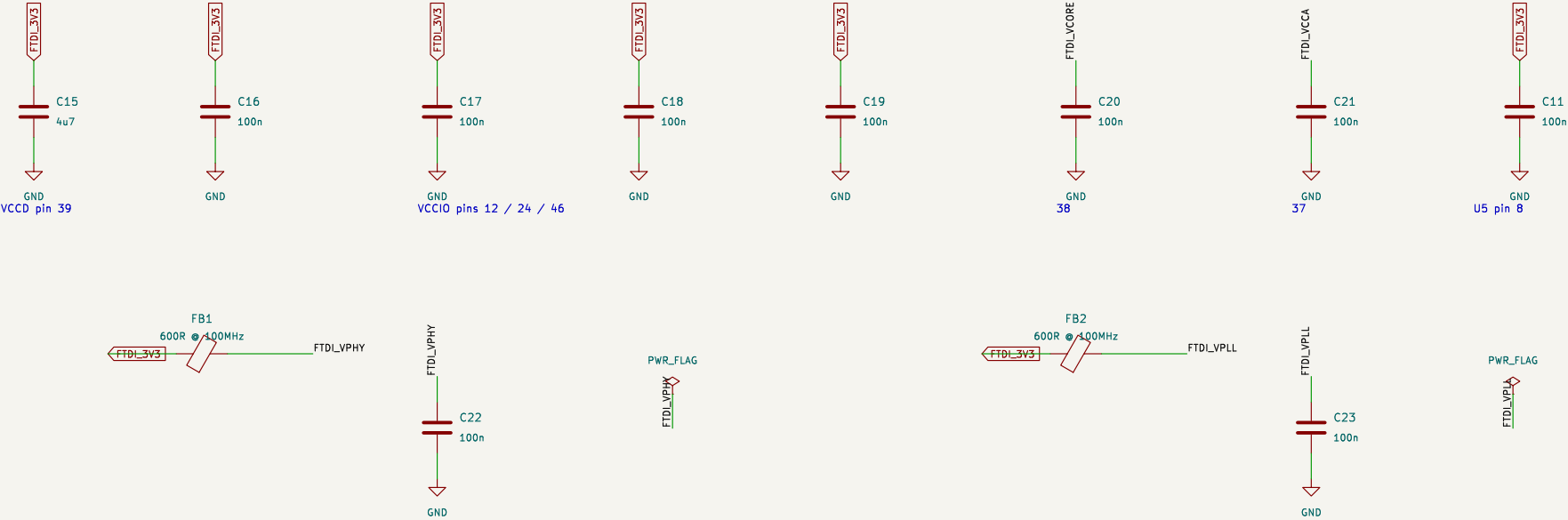


Unused ADBUS/ACBUS pins: no connection.
ACBUS5/6 recovery deferred to Rev B.
VCCA and VCORE each bypass only; never join.

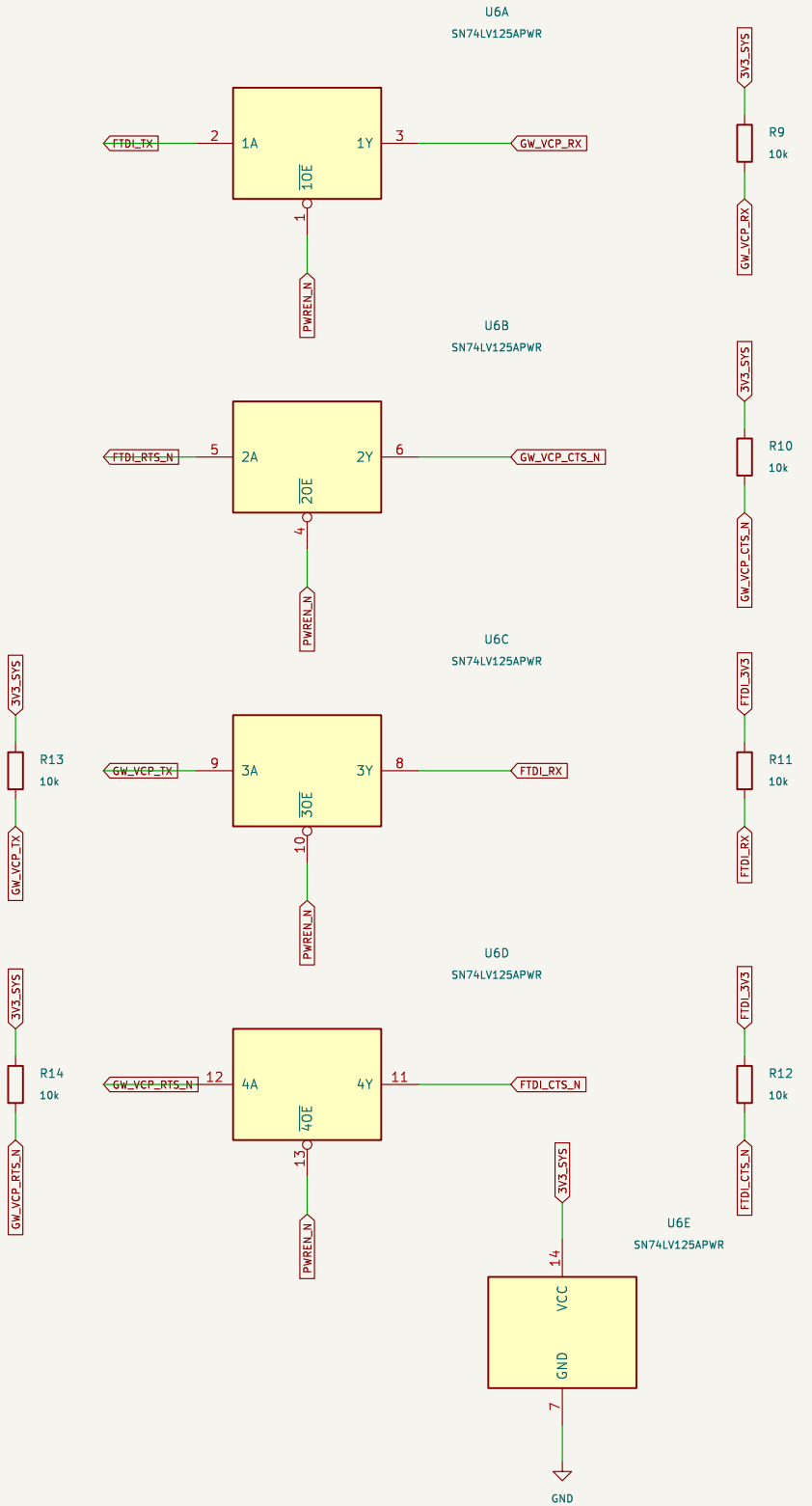
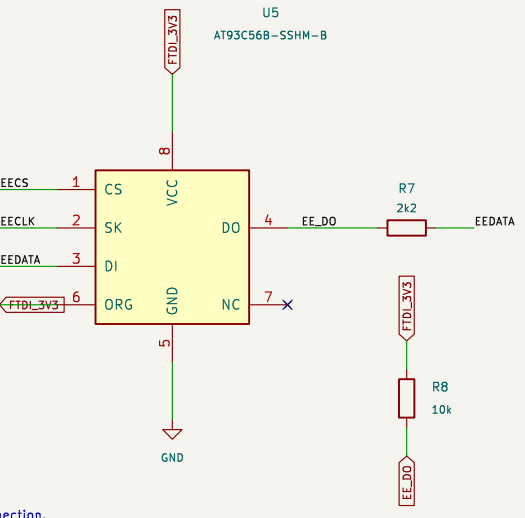
ORG=VCC: 128 x 16 words.
EEDATA to Di: DO through 2k2.
10k pull-up belongs on DO side.

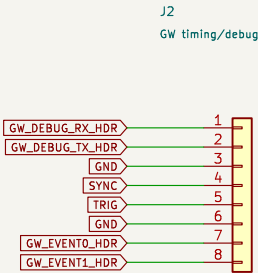
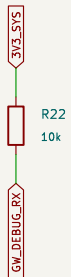
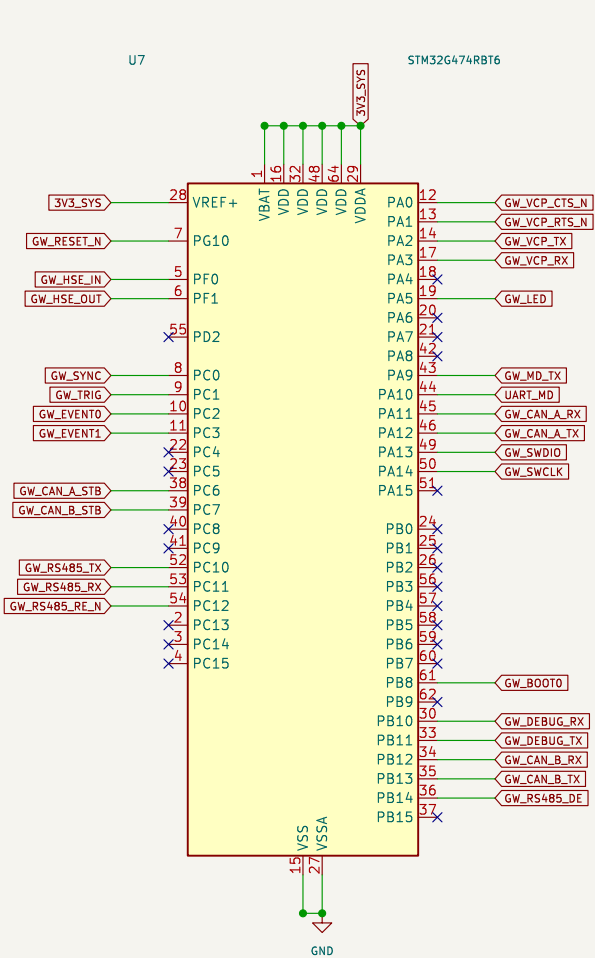
GW-* directions are from the MCU. TX/RX idle high; CTS_N inactive high.
All /OE pins directly use PWREN_N (ADR-040).
These four GW nets connect to the gateway MCU sheet.

FTDI LOCAL SUPPLY DECOUPLING



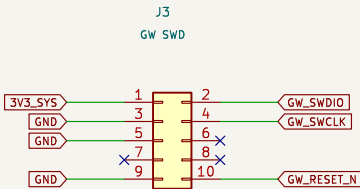
Separate ferrite + 100n at VPHY pin 3 and VPLL pin 8.
FTDI Fig.6.1 optional post-bed bulk not fitted; no DNP footprints.
All AGND/GND pins join the board ground plane.





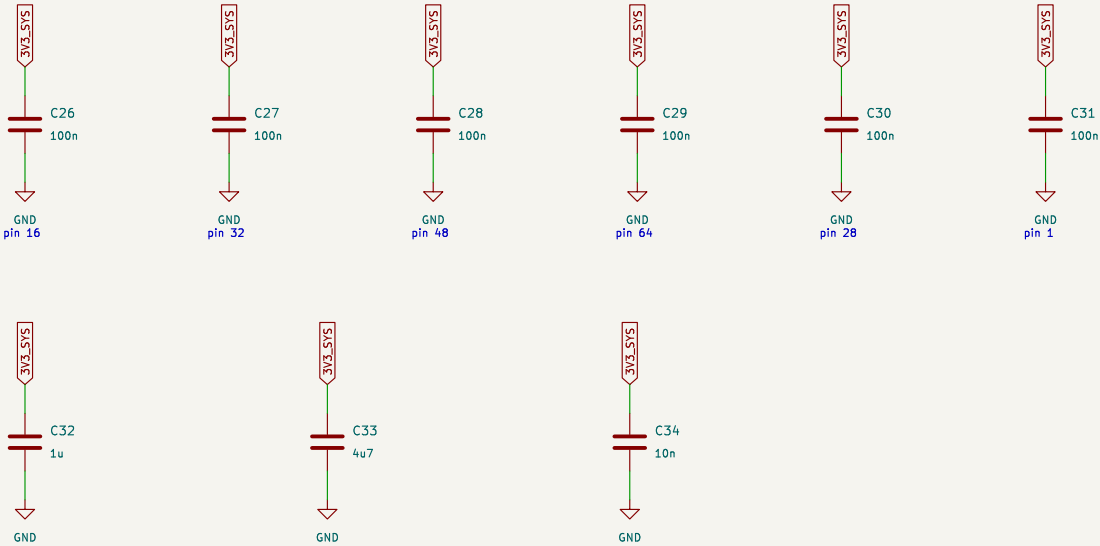
330R at each MCU pad. Header is 3.3 V logic.
RX/TX names use the MCU perspective.
One external source per global SYNC / TRIG.
Drive only while SYS is on; disconnect before suspend.

INDEPENDENT SWD

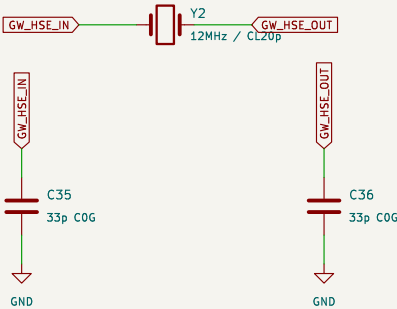


Vtref is sense only; probe must not power target.
6 SWD / 8 TDI unused; 7 is NC / key position.
Confirm cable key before insertion. Reset on pin 10.

LOCAL SUPPLIES



1u shared VDDA/VREF+ bulk; 10n at VDDA pin29.
VREFBUF disabled / high-Z; VBAT tied to main rail.
Retain NRST function / normal Flash boot option bytes.



Rev A complete schematic draft – no PCB layout

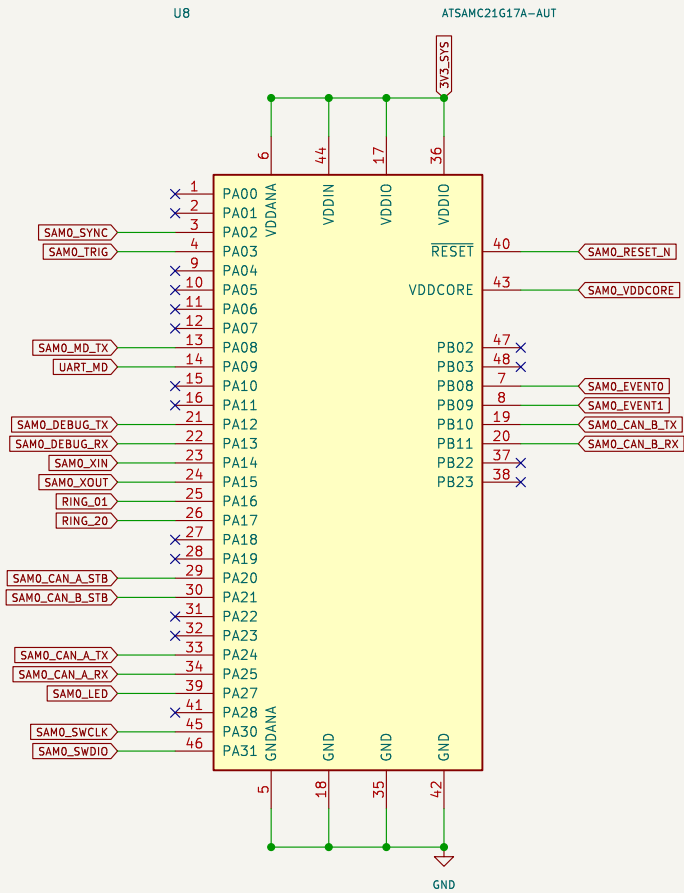
WireSpaces

Sheet: /Gateway STM32G474/
File: gateway.kicad_sch

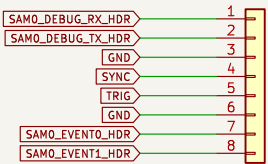
Title: Board 1 – GW MCU, debug and timing

Size: A3 Date: 2026-09-07
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: A-draft
Id: 4/11



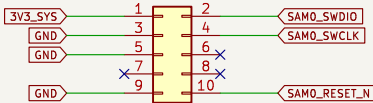
J4
SAM0 timing/debug



330R at each MCU pad. Header is 3.3 V logic.
RX/TX names use the MCU perspective.
One external source per global SYNC / TRIG.
Drive only while SYS is on; disconnect before suspend.

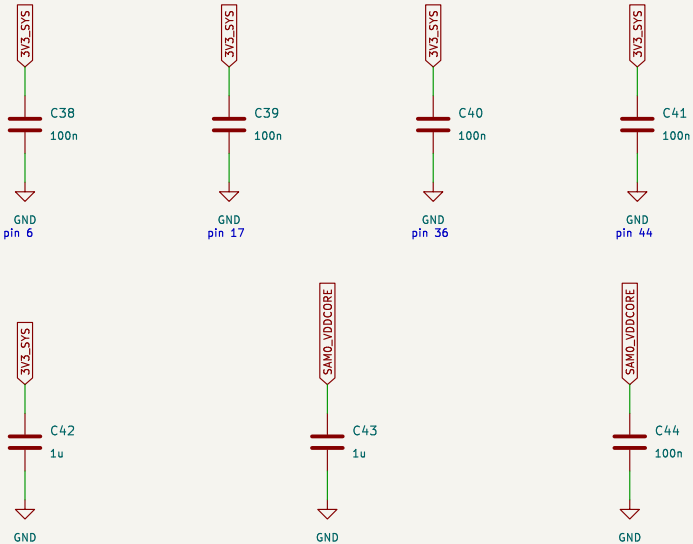
INDEPENDENT SWD

J5
SAM0 SWD

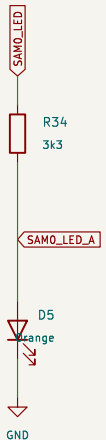
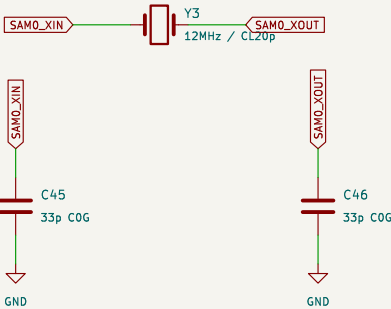


Vtref is sense only; probe must not power target.
6 SWD / 8 TDI unused; 7 is NC / key position.
Confirm cable key before insertion. Reset on pin 10.

LOCAL SUPPLIES



1u main bulk; separate VDDCORE output capacitors.
Never join core rails or connect them to 3V3_SYS.
EIC: synchronous mode; clear flags after enabling.



Rev A complete schematic draft – no PCB layout

WireSpaces

Sheet: /SAM0/

File: sam0.kicad_sch

Title: Board 1 – SAM0 MCU, debug and timing

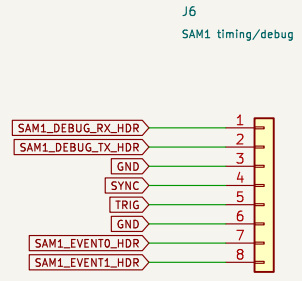
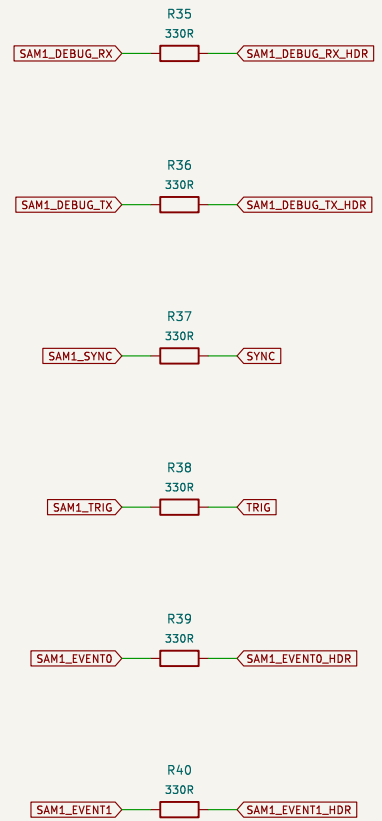
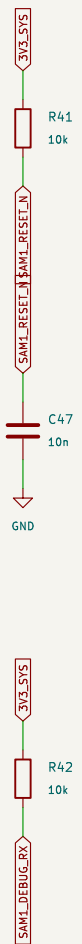
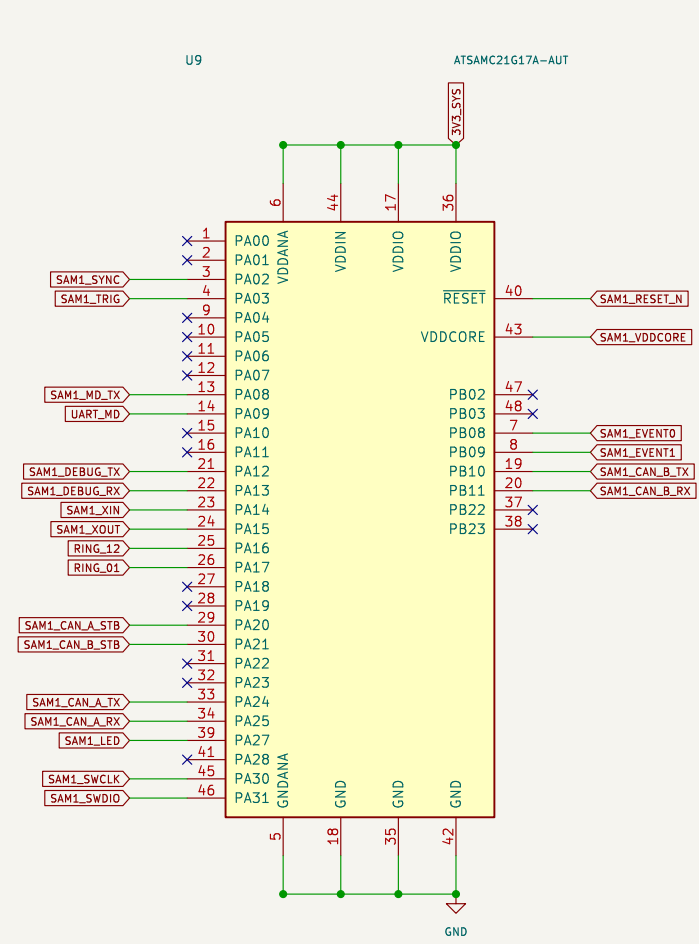
Size: A3

Date: 2026-09-07

Rev: A-draft

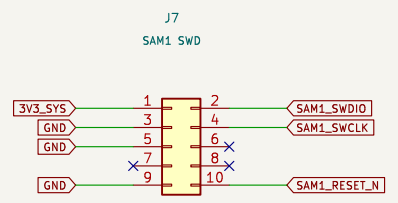
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 5/11



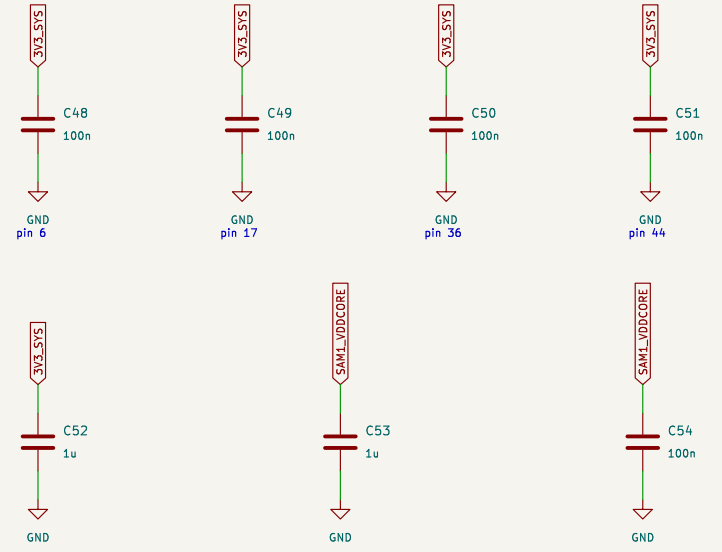
330R at each MCU pad. Header is 3.3 V logic.
RX/TX names use the MCU perspective.
One external source per global SYNC / TRIG.
Drive only while SYS is on; disconnect before suspend.

INDEPENDENT SWD

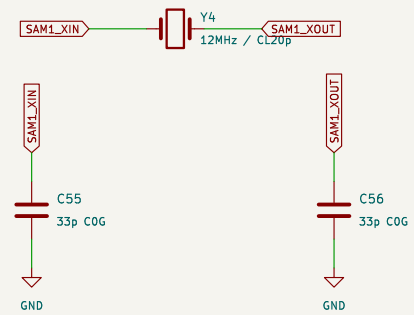
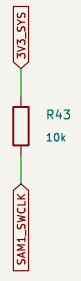


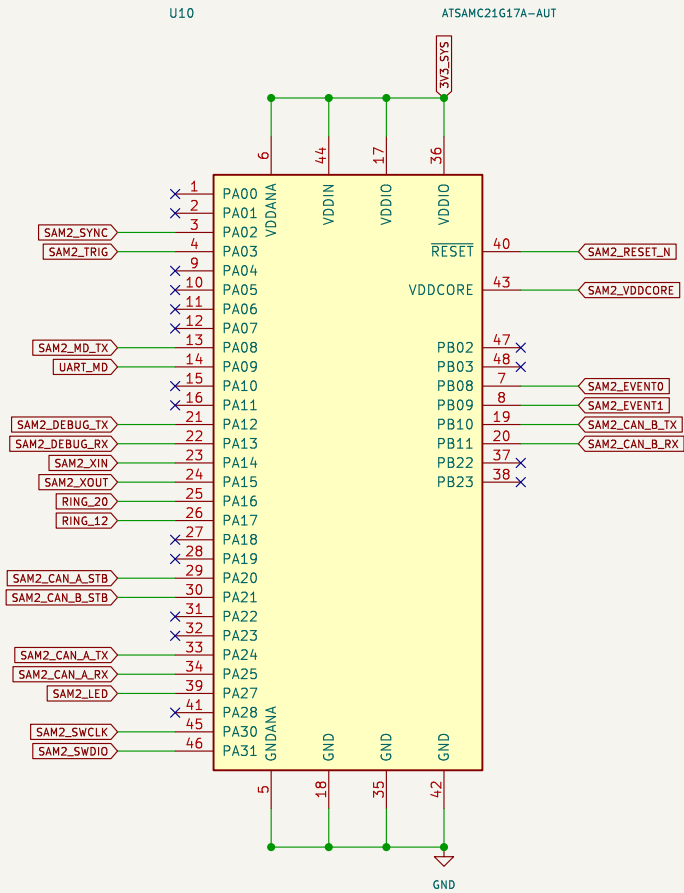
Vtref is sense only; probe must not power target.
6 SWD / 8 TDI unused; 7 is NC / key position.
Confirm cable key before insertion. Reset on pin 10.

LOCAL SUPPLIES

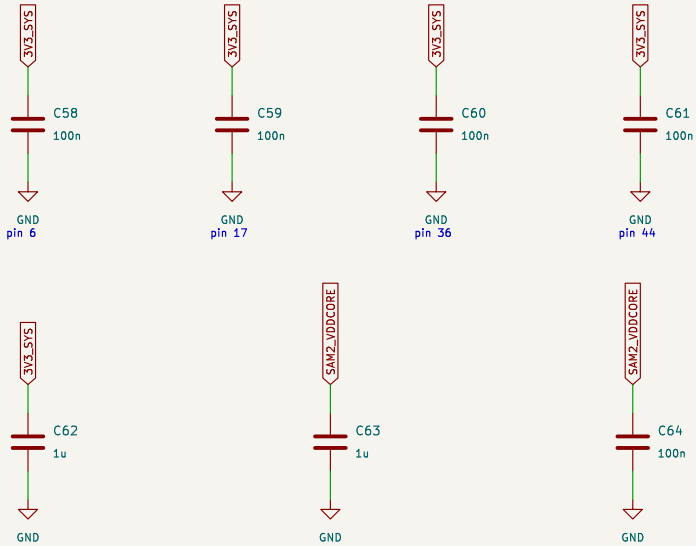


1u main bulk; separate VDDCORE output capacitors.
Never join core rails or connect them to 3V3_SYS.
EIC: synchronous mode; clear flags after enabling.

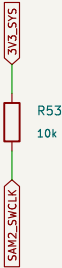




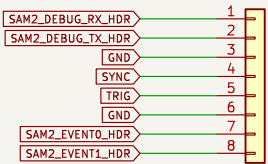
LOCAL SUPPLIES



1u main bulk; separate VDDCORE output capacitors.
Never join core rails or connect them to 3V3_SYS.
EIC: synchronous mode; clear flags after enabling.



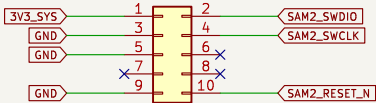
J8
SAM2 timing/debug



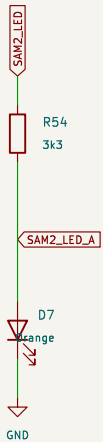
330R at each MCU pad. Header is 3.3 V logic.
RX/TX names use the MCU perspective.
One external source per global SYNC / TRIG.
Drive only while SYS is on; disconnect before suspend.

INDEPENDENT SWD

J9
SAM2 SWD



Vtref is sense only; probe must not power target.
6 SWD / 8 TDI unused; 7 is NC / key position.
Confirm cable key before insertion. Reset on pin 10.



Rev A complete schematic draft – no PCB layout

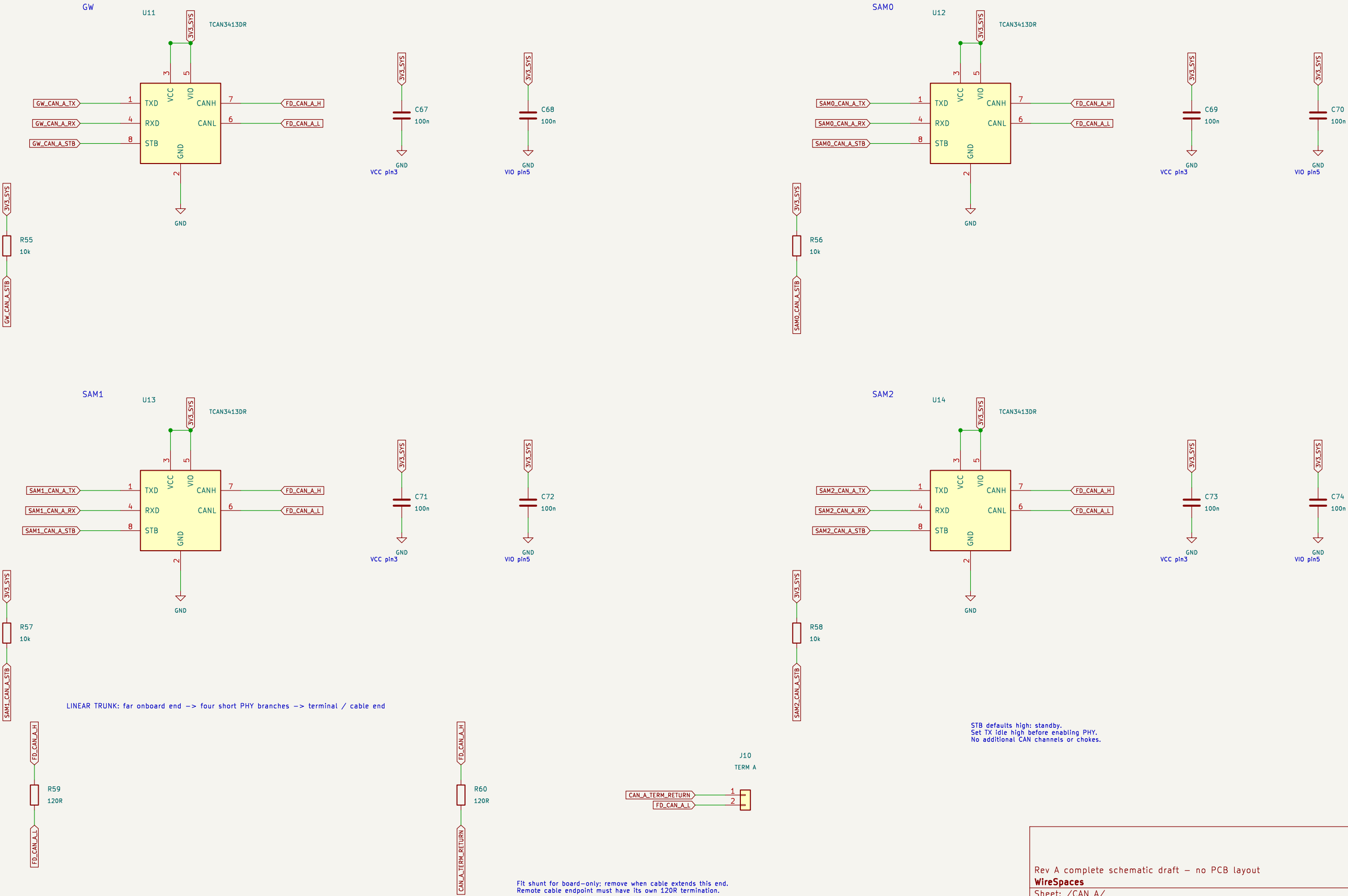
WireSpaces

Sheet: /SAM2/
File: sam2.kicad_sch

Title: Board 1 – SAM2 MCU, debug and timing

Size: A3 Date: 2026-09-07
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: A-draft
Id: 7/11



Rev A complete schematic draft - no PCB layout

WireSpaces

Sheet: /CAN A/

File: can_a.kicad_sch

Title: Board 1 - CAN A / four independent PHYs

Size: A3

Date: 2026-09-07

Rev: A-draft

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 8/11

